Lesson 12: Repeated Games and Incomplete Information

Folk Theorems, Bayesian Games, and Potential Games

Session 12 of 102 | Phase I: Mathematical Foundations | 8 h
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Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

Abstract

This document develops the theory of repeated games, games of incomplete information, and potential games, with complete proofs of all major results. We establish the folk theorems—characterizing the equilibrium payoff sets of infinitely repeated games—prove convergence of fictitious play and no-regret dynamics in zero-sum games, develop the Bayesian game framework and perfect Bayesian equilibrium for signaling games, and prove that potential games admit pure Nash equilibria with convergent best-response dynamics. Throughout, we connect these classical results to multi-agent reinforcement learning: independent learners in repeated games, opponent modeling via Bayesian reasoning, self-play algorithms, and convergence guarantees in potential games.

Prerequisites: Normal-form games, Nash equilibrium, mixed strategies, minimax theorem (Lesson 11).

Notation: We write $[n] = \{1, \ldots, n\}$ for player sets. For a strategy profile $a = (a_1, \ldots, a_n)$, we write $a_{-i} = (a_1, \ldots, a_{i-1}, a_{i+1}, \ldots, a_n)$ and (a_i', a_{-i}) for the profile where player i deviates to a_i'. Stage-game payoffs are $g_i(a)$; repeated-game payoffs are u_i.

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1 Introduction

In Lesson 11 we studied one-shot (normal-form) games: simultaneous moves, Nash equilibrium, and the minimax theorem. However, most real strategic interactions—and virtually all multi-agent reinforcement learning (MARL) scenarios—involve *repeated interaction*. Agents meet the same opponents across many episodes, observe each other's past behavior, and adapt their strategies over time. Three fundamental phenomena emerge:

(i) **Cooperation through repetition.** Even when one-shot incentives favor defection (as in the Prisoner's Dilemma), the threat of future punishment can sustain cooperative outcomes. The *folk theorems* characterize exactly which payoff vectors are achievable.

(ii) **Incomplete information.** In practice, agents rarely know each other's payoff functions. A robot negotiating with a human does not know the human's utility; competing firms do not know rivals' costs. *Bayesian games* model this uncertainty, and *perfect Bayesian equilibrium* handles sequential decision-making under such uncertainty.

(iii) **Structured interactions.** Many multi-agent problems possess special structure—such as *potential games*, where a single global function captures all players' incentives. In these settings, simple learning rules (best-response dynamics, log-linear learning) are guaranteed to converge.

This document covers all three themes with complete proofs. The roadmap is as follows.

- **Section 2**: Finitely repeated games and the unraveling argument.
- **Section 3**: Infinitely repeated games, trigger strategies, and feasible/individually rational payoffs.
- **Section 4**: The Nash folk theorem and the subgame perfect folk theorem.
- **Section 5**: Fictitious play, no-regret learning, and convergence in zero-sum games.
- **Section 6**: Bayesian games and Bayesian Nash equilibrium.
- **Section 7**: Perfect Bayesian equilibrium and signaling games.
- **Section 8**: Potential games, congestion games, and convergence of best-response dynamics.
- **Section 9**: Connections to multi-agent RL.
- **Section 10**: Exercises.
- **Section 11**: Summary of key results.

Why This Matters for MARL

In multi-agent RL, agents interact repeatedly and update their policies based on observed outcomes. The folk theorems explain why independent Q-learners can converge to cooperative (or adversarial) outcomes that do not arise in single-episode analysis. Bayesian games formalize the partial observability inherent in opponent modeling. Potential games identify problem classes where independent learning is guaranteed to converge—a rare and valuable property in general-sum MARL.

2 Finitely Repeated Games

2.1 Model

We begin with a *stage game* $G = (N, (A_i)_{i\in N}, (g_i)_{i\in N})$ where $N = [n]$ is the player set, A_i is the finite action set of player i, and $g_i : A \to \mathbb{R}$ is the stage-game payoff function, with $A = \prod_{i\in N} A_i$.

Definition 2.1 (Finitely Repeated Game)**.** The *T-fold repetition* of G, denoted G^T, is defined as follows:

(a) In each period $t \in \{1, 2, \ldots, T\}$, all players simultaneously choose actions $a_i^t \in A_i$.

(b) At the end of period t, all players observe the realized action profile $a^t = (a_1^t, \ldots, a_n^t)$.

(c) The *history* at period t is $h^t = (a^1, a^2, \ldots, a^{t-1}) \in A^{t-1}$. The initial history is $h^1 = \emptyset$.

(d) A *(pure) strategy* for player i is a sequence of functions $s_i = (s_i^1, s_i^2, \ldots, s_i^T)$ where $s_i^t : A^{t-1} \to A_i$ maps histories to actions.

(e) The payoff to player i under strategy profile $s = (s_1, \ldots, s_n)$ is

$$u_i(s) = \frac{1}{T}\sum_{t=1}^{T} g_i\big(a^t(s, h^t)\big),$$

where $a^t(s, h^t) = \big(s_1^t(h^t), \ldots, s_n^t(h^t)\big)$ and histories are generated by the strategy profile.

Remark 2.1. We use average payoffs $\frac{1}{T}\sum_t g_i(a^t)$ rather than total payoffs to keep the repeated-game payoffs on the same scale as stage-game payoffs, facilitating comparison with the infinite-horizon discounted case.

2.2 Subgame Perfect Equilibrium

The repeated game G^T is an extensive-form game of perfect information (after each period, all actions are observed). Thus the natural solution concept is *subgame perfect equilibrium* (SPE): a strategy profile that constitutes a Nash equilibrium in every subgame.

Definition 2.2 (Subgame Perfect Equilibrium)**.** A strategy profile s^* in G^T is a *subgame perfect equilibrium* (SPE) if, for every period $t \in \{1, \ldots, T\}$ and every history $h^t \in A^{t-1}$, the continuation strategy profile $(s_i^{*,\tau}(h^t, \cdot))_{\tau\geq t}$ is a Nash equilibrium of the subgame starting from h^t.

2.3 The Unraveling Argument

The fundamental result for finitely repeated games with a unique stage-game equilibrium is that repetition adds nothing: the unique SPE plays the stage-game NE in every period.

Theorem 2.1 (Unique NE Implies Unique SPE)**.** *If the stage game G has a unique Nash equilibrium $a^* = (a_1^*, \ldots, a_n^*)$, then the T-fold repeated game G^T has a unique subgame perfect equilibrium in which $a^t = a^*$ for all $t \in \{1, \ldots, T\}$.*

Proof. We proceed by backward induction on t.

Base case ($t = T$): In the final period, regardless of history h^T, the subgame is identical to the one-shot stage game G. Since a^* is the unique NE of G, every SPE must play a^* in period T.

Inductive step: Suppose that in every SPE, $a^\tau = a^*$ for all $\tau \in \{t + 1, \ldots, T\}$, regardless of the history at period $t + 1$. Consider the subgame beginning at an arbitrary history h^t.

The continuation payoffs from period $t+1$ onward are fixed at $\frac{1}{T}\sum_{\tau=t+1}^{T} g_i(a^*) = \frac{T-t}{T} g_i(a^*)$, independent of the action chosen at period t (since all future play is a^* regardless of history). Therefore, player i's effective payoff at period t is

$$\frac{1}{T} g_i(a^t) + \frac{T-t}{T} g_i(a^*),$$

and maximizing this over a_i^t is equivalent to maximizing $g_i(a_i^t, a_{-i}^t)$ in the stage game. Since a^* is the unique NE, $a^t = a^*$ in every SPE.

By induction, $a^t = a^*$ for all t, proving that the unique SPE is to play a^* in every period. □

The Unraveling Trap

Theorem 2.1 shows that finite repetition of the Prisoner's Dilemma (which has a unique NE at mutual defection) cannot sustain cooperation under subgame perfection. This is counterintuitive and motivates the study of infinitely repeated games, where the "last period" argument cannot be applied.

2.4 Multiple Stage-Game Equilibria

When the stage game has multiple Nash equilibria, repetition *can* create new equilibrium outcomes.

Proposition 2.2 (New Equilibria Through Repetition)**.** *If the stage game G has at least two Nash equilibria yielding different payoffs for some player, then the two-fold repeated game G^2 may have SPE outcomes in which first-period play is not a stage-game NE.*

Proof. We give a constructive proof. Let G have two Nash equilibria a^* and $\hat{a}$ with $g_i(a^*) > g_i(\hat{a})$ for all i (i.e., a^* Pareto-dominates $\hat{a}$). Suppose there exists an action profile $\tilde{a}$ (not necessarily a NE) with $g_i(\tilde{a}) > g_i(a^*)$ for all i, and suppose the gain from deviating unilaterally in the stage game satisfies

$$\max_{a_i' \in A_i} g_i(a_i', \tilde{a}_{-i}) - g_i(\tilde{a}) < g_i(a^*) - g_i(\hat{a}).$$

Consider the following strategy for each player i in G^2:

- Period 1: Play $\tilde{a}_i$.
- Period 2: If $a^1 = \tilde{a}$, play a_i^*. Otherwise, play $\hat{a}_i$.

In period 2, players play a stage-game NE (a^* or $\hat{a}$) depending on period-1 history, so the strategy is sequentially rational in period 2.

In period 1, if player i deviates to a_i' while others play $\tilde{a}_{-i}$, the payoff change is:

$$\begin{aligned}&\frac{1}{2}\big[g_i(a_i', \tilde{a}_{-i}) + g_i(\hat{a})\big] - \frac{1}{2}\big[g_i(\tilde{a}) + g_i(a^*)\big]\\ &= \frac{1}{2}\big[g_i(a_i', \tilde{a}_{-i}) - g_i(\tilde{a})\big] - \frac{1}{2}\big[g_i(a^*) - g_i(\hat{a})\big] < 0,\end{aligned}$$

by the assumed inequality. Thus no player wants to deviate, and the profile is an SPE in which first-period play $\tilde{a}$ is *not* a stage-game NE. □

Example 2.1 (Coordination Game)**.** Consider the two-player coordination game with payoffs:

	L	R
T	$3,3$	$0,0$
B	$0,0$	$1,1$

This has two pure NE: (T, L) with payoffs $(3,3)$ and (B, R) with payoffs $(1,1)$. In the two-fold repetition, the "punishment" equilibrium (B, R) can incentivize cooperation on outcomes that are not stage-game NE, provided the incentive condition is satisfied.

3 Infinitely Repeated Games

3.1 Model

To circumvent the unraveling argument, we consider infinite repetition with discounting.

Definition 3.1 (Infinitely Repeated Game)**.** Given a stage game G and a discount factor $\delta \in (0,1)$, the *infinitely repeated game* $G^\infty(\delta)$ is defined by:

(a) In each period $t \in \{1, 2, 3, \ldots\}$, players simultaneously choose actions $a^t \in A$.

(b) Histories, strategies are defined as in the finite case, but with t ranging over $\mathbb{N}$.

(c) Player i's payoff under strategy profile s is the *normalized discounted sum*:

$$u_i(s) = (1-\delta)\sum_{t=1}^{\infty} \delta^{t-1} g_i(a^t).$$

The normalization factor $(1-\delta)$ ensures that payoffs lie in the same range as stage-game payoffs: if $a^t = a$ for all t, then $u_i = g_i(a)$.

3.2 Feasible and Individually Rational Payoffs

Definition 3.2 (Feasible Payoff Set)**.** The *feasible payoff set* of the stage game G is

$$V = \operatorname{conv}\{g(a) : a \in A\} = \operatorname{conv}\{(g_1(a), \ldots, g_n(a)) : a \in A\} \subset \mathbb{R}^n,$$

the convex hull of all achievable stage-game payoff vectors.

Remark 3.1. In the infinitely repeated game, any payoff vector $v \in V$ is achievable by cycling through action profiles with appropriate frequencies. For instance, if $v = \lambda g(a) + (1-\lambda) g(a')$ for some $\lambda \in [0,1]$, then playing a for a fraction λ of periods and a' for the remaining fraction achieves payoff v in the limit as $\delta \to 1$.

Definition 3.3 (Minimax Value and Individually Rational Payoffs)**.** The *minimax value* (or *minmax value*) of player i in pure strategies is

$$\underline{v}_i = \min_{a_{-i} \in A_{-i}} \max_{a_i \in A_i} g_i(a_i, a_{-i}).$$

The *mixed-strategy minimax value* is

$$\underline{v}_i^{\text{mix}} = \min_{\alpha_{-i} \in \Delta(A_{-i})} \max_{a_i \in A_i} \mathbb{E}_{\alpha_{-i}}[g_i(a_i, a_{-i})],$$

where $\Delta(A_{-i}) = \prod_{j \neq i} \Delta(A_j)$ is the set of independent mixed strategy profiles for i's opponents. A payoff vector $v \in \mathbb{R}^n$ is *individually rational* if $v_i \geq \underline{v}_i^{\text{mix}}$ for all i.

Proposition 3.1. *In any Nash equilibrium of $G^\infty(\delta)$, each player i receives payoff $u_i \geq \underline{v}_i^{mix}$.*

Proof. In any NE, player i can deviate to the strategy of playing $\operatorname{argmax}_{a_i} \mathbb{E}_{\alpha_{-i}^t}[g_i(a_i, a_{-i}^t)]$ in every period t (where α_{-i}^t is the marginal distribution of opponents' play at time t induced by the equilibrium). This guarantees a stage payoff of at least $\underline{v}_i^{\text{mix}}$ in every period, hence a total payoff of at least $\underline{v}_i^{\text{mix}}$. Since the NE strategy must yield payoff at least as high as any deviation, we get $u_i \geq \underline{v}_i^{\text{mix}}$. □

3.3 Trigger Strategies

Definition 3.4 (Grim Trigger Strategy)**.** Given a target action profile $a^* \in A$, the *grim trigger strategy* for player i is:

$$s_i^t(h^t) = \begin{cases} a_i^* & \text{if } a^\tau = a^* \text{ for all } \tau < t, \\ m_i & \text{otherwise,} \end{cases}$$

where $m_i \in \operatorname{argmin}_{\alpha_i} \max_{a_j} g_j(a_j, \alpha_{-j})$ is player i's component of a minimax punishment profile against any deviator.

Example 3.1 (Cooperation in the Repeated Prisoner's Dilemma)**.** Consider the Prisoner's Dilemma with payoffs:

	C	D
C	$3,3$	$0,4$
D	$4,0$	$1,1$

The unique stage-game NE is (D, D) with payoffs $(1, 1)$. The minimax value for each player is $\underline{v}_i = 1$. Consider the grim trigger strategy profile that plays (C, C) and punishes with (D, D) forever.

Cooperation payoff: If both cooperate forever, each gets $u_i = 3$.

Deviation payoff: If player i defects in period t (while the opponent cooperates), she gets 4 in that period and 1 thereafter:

$$u_i^{\text{dev}} = (1-\delta)\big[\delta^{t-1} \cdot 4 + \sum_{\tau=t+1}^{\infty} \delta^{\tau-1} \cdot 1\big] + (1-\delta) \sum_{\tau=1}^{t-1} \delta^{\tau-1} \cdot 3.$$

The best deviation is at $t = 1$:

$$u_i^{\text{dev}} = (1-\delta) \cdot 4 + \delta \cdot 1 = 4 - 3\delta.$$

Cooperation is sustained iff $3 \geq 4 - 3\delta$, i.e., $\delta \geq \frac{1}{3}$.

Discount Factors and Episode Length in RL

The discount factor δ in repeated games is analogous to the discount factor γ in MDPs. In MARL, if γ is too small (agents are too myopic), cooperative equilibria may not be sustainable. This directly parallels the threshold $\delta \geq 1/3$ in the repeated Prisoner's Dilemma: myopic agents (small γ) cannot learn to cooperate.

Definition 3.5 (Tit-for-Tat)**.** In a two-player game, the *tit-for-tat* strategy for player i is:

$$s_i^1 = C, \qquad s_i^t(h^t) = a_j^{t-1} \quad \text{for } t \geq 2,$$

where $j \neq i$. That is, start by cooperating, then mimic the opponent's previous action.

Remark 3.2. Tit-for-tat is a "forgiving" strategy: unlike grim trigger, punishment is temporary. Axelrod's famous tournaments showed that tit-for-tat performs well in evolutionary settings, which connects to the multi-agent RL idea that simple, interpretable strategies can be surprisingly effective.

4 Folk Theorems

The folk theorems characterize the equilibrium payoff set of infinitely repeated games. Informally, "almost anything" can be sustained as an equilibrium if players are patient enough.

4.1 Nash Folk Theorem

Theorem 4.1 (Nash Folk Theorem). *Let G be a finite normal-form game and $V = \text{conv}\{g(a) : a \in A\}$ the feasible payoff set. For any payoff vector $v \in V$ satisfying $v_i > \underline{v}_i^{mix}$ for all i (strictly individually rational), there exists $\underline{\delta} \in (0,1)$ such that for all $\delta \in (\underline{\delta}, 1)$, there is a Nash equilibrium of $G^\infty(\delta)$ with payoff vector v.*

Proof. We construct a Nash equilibrium yielding payoff v.

Step 1: Achieving v through action sequences. Since $v \in V = \text{conv}\{g(a) : a \in A\}$, there exist action profiles $a(1), a(2), \ldots, a(K) \in A$ and weights $\lambda_1, \ldots, \lambda_K \geq 0$ with $\sum_k \lambda_k = 1$ such that $v = \sum_k \lambda_k g(a(k))$. We can approximate v by cycling through these action profiles. Specifically, for any $\varepsilon > 0$ and δ close enough to 1, we can construct a deterministic sequence of action profiles $(a^t)_{t=1}^\infty$ such that the discounted average payoff $(1-\delta)\sum_{t=1}^\infty \delta^{t-1} g(a^t)$ is within ε of v.

More precisely, define a cycle of length L where action profile $a(k)$ is played for $\lceil \lambda_k L \rceil$ periods (adjusting the last to make the total exactly L). As $\delta \to 1$, the discounted average of any periodic sequence converges to the time-average, which converges to v as $L \to \infty$. Fix $\varepsilon > 0$ small enough that $v_i - \varepsilon > \underline{v}_i^{\text{mix}}$ for all i, and choose L and $\underline{\delta}_1$ accordingly.

Step 2: Punishment strategies. For each player i, let $\alpha_{-i}^{m,i} \in \Delta(A_{-i})$ be a mixed minimax punishment profile: $\max_{a_i} \mathbb{E}_{\alpha_{-i}^{m,i}}[g_i(a_i, a_{-i})] = \underline{v}_i^{\text{mix}}$. In the punishment phase, players $j \neq i$ play according to $\alpha_{-i}^{m,i}$ independently.

Step 3: Strategy construction. Define the following strategy for each player j:

- *Cooperation phase:* Play the sequence $(a^t)_{t=1}^\infty$ from Step 1.
- *Punishment phase:* If some player i is the first to deviate from the prescribed sequence in some period $\hat{t}$, then for all $t > \hat{t}$, all players $j \neq i$ play according to $\alpha_{-i}^{m,i}$ (punishing i forever).

Step 4: Verifying the Nash equilibrium condition. Consider whether player i wants to deviate. If player i follows the prescribed strategy, she receives payoff within ε of v_i (for $\delta > \underline{\delta}_1$).

If player i deviates at any period $\hat{t}$, she receives at most $\max_a g_i(a) \cdot (1-\delta)\delta^{\hat{t}-1}$ in that period, plus the punishment continuation payoff. In the punishment phase, the opponents play the mixed minimax strategy, so player i's best response yields at most $\underline{v}_i^{\text{mix}}$ per period. Thus the deviation payoff is at most:

$$u_i^{\text{dev}} \leq (1-\delta)\sum_{\tau=1}^{\hat{t}} \delta^{\tau-1} M + \delta^{\hat{t}} \underline{v}_i^{\text{mix}}$$

where $M = \max_a |g_i(a)|$.

As $\delta \to 1$, the first term (a finite number of periods) vanishes and $u_i^{\text{dev}} \to \underline{v}_i^{\text{mix}} < v_i - \varepsilon$. Therefore, for δ large enough (say $\delta > \underline{\delta}_2$), deviation is unprofitable.

Taking $\underline{\delta} = \max(\underline{\delta}_1, \underline{\delta}_2)$ completes the proof.

Note on Nash vs. SPE: This equilibrium is Nash but not necessarily subgame perfect: punishing players have no incentive to carry out the punishment (they are indifferent or may prefer not to punish). The subgame perfect folk theorem below addresses this. □

Nash Folk Theorem – Interpretation

Any feasible, strictly individually rational payoff vector is achievable as a Nash equilibrium payoff for sufficiently patient players. This means the set of NE payoffs of the repeated game "fills up" the feasible individually rational region as $\delta \to 1$.

4.2 Subgame Perfect Folk Theorem

The Nash folk theorem has a weakness: punishment threats may not be credible. The punishers might prefer to deviate from punishment. The subgame perfect folk theorem (Fudenberg and Maskin, 1986) resolves this by constructing self-enforcing punishment schemes.

Definition 4.1 (Full Dimensionality Condition)**.** The stage game G satisfies the *full dimensionality condition* if the feasible payoff set $V = \text{conv}\{g(a) : a \in A\}$ has dimension n (the number of players), i.e., V has nonempty interior in $\mathbb{R}^n$.

Theorem 4.2 (Subgame Perfect Folk Theorem (Fudenberg–Maskin))**.** *Let G be a finite n-player game satisfying the full dimensionality condition. For any feasible payoff vector $v \in V$ with $v_i > \underline{v}_i^{mix}$ for all i, there exists $\underline{\delta} \in (0,1)$ such that for all $\delta \in (\underline{\delta}, 1)$, there is a subgame perfect equilibrium of $G^\infty(\delta)$ with payoff vector v.*

Proof. The proof is constructive. We build a strategy profile with multiple phases: a cooperation phase and player-specific punishment phases, arranged so that each phase is self-enforcing.

Step 1: Setup. Fix $v \in V$ with $v_i > \underline{v}_i^{\text{mix}}$ for all i. Since V has full dimension, for each player i there exists a feasible payoff vector $w^i \in V$ with:

(i) $w_j^i > \underline{v}_j^{\text{mix}}$ for all $j \neq i$ (the punishers receive strictly above their minimax values during the punishment of i),

(ii) $w_i^i < v_i$ (player i's payoff during her own punishment phase is below her cooperative payoff; we can choose w_i^i close to $\underline{v}_i^{\text{mix}}$).

The full dimensionality condition ensures such w^i exist: since V has nonempty interior, we can find vectors that independently adjust each player's payoff.

Let $m_{-i}^i \in \Delta(A_{-i})$ be a mixed minimax strategy profile against player i.

Step 2: Strategy construction (three-phase strategy).

Phase 0 (Cooperation): Play the action sequence that yields payoff v (as in the Nash folk theorem proof).

Phase I_i (Punishment of player i): Lasts for L periods. During this phase, the opponents of i play the minimax strategy m_{-i}^i in each period, and player i plays any best response (which yields at most $\underline{v}_i^{\text{mix}}$ per period).

Phase II_i (Reward after punishment): After the L-period punishment phase, play switches to the action sequence yielding payoff w^i forever. This rewards the punishers (who received above their minimax during punishment) and keeps i's continuation payoff below v_i if L is large enough.

Transitions:

- If player j is the first to deviate in Phase 0, switch to Phase I_j.
- If player j deviates during Phase I_i (and $j \neq i$), restart Phase I_j (punish the deviating punisher).
- If player i deviates during her own punishment Phase I_i, restart Phase I_i.

Step 3: Choosing L. We need L large enough that deviation is unprofitable. Let $M = \max_{a,i} |g_i(a)|$. During Phase I_i, player i gets at most $\underline{v}_i^{\text{mix}}$ per period. After Phase I_i, player i gets w_i^i per period. Thus player i's continuation payoff from Phase I_i is:

$$(1-\delta)\sum_{t=1}^{L} \delta^{t-1}\underline{v}_i^{\text{mix}} + \delta^L w_i^i = (1-\delta^L)\underline{v}_i^{\text{mix}} + \delta^L w_i^i.$$

We need this to be less than v_i (so punishment deters deviation from Phase 0) and for the punishers' payoffs to make Phase I self-enforcing.

Step 4: Verifying sequential rationality.

Phase 0 – Player i considers deviating: Deviation yields at most M in one period, then Phase I_i begins. Compliance yields v_i forever. The deviation is unprofitable when:

$$v_i \geq (1-\delta)M + \delta\big[(1-\delta^L)\underline{v}_i^{\text{mix}} + \delta^L w_i^i\big].$$

As $\delta \to 1$, the right side approaches $(1-\delta)M + \delta[(1-\delta^L)\underline{v}_i^{\text{mix}} + \delta^L w_i^i]$. For δ close to 1 and L large, $(1-\delta)M \to 0$ and the continuation approaches a weighted average of $\underline{v}_i^{\text{mix}}$ and w_i^i, both strictly below v_i (since $w_i^i < v_i$ and $\underline{v}_i^{\text{mix}} < v_i$). So the inequality holds for δ large enough.

Phase I_i – Punisher $j \neq i$ considers deviating: If j deviates during Phase I_i, Phase I_j begins (punishing j). Compliance with Phase I_i yields: the minimax strategy payoff for $L-t$ remaining periods (at worst $-M$ per period but bounded), followed by $w_j^i > \underline{v}_j^{\text{mix}}$ forever. Deviation yields at most M for one period, then Phase I_j punishment.

For δ close to 1, the continuation payoff from compliance converges to $w_j^i > \underline{v}_j^{\text{mix}}$, while deviation leads eventually to w_j^j which can be chosen close to $\underline{v}_j^{\text{mix}}$. So compliance dominates for large δ.

Phase I_i – Player i considers deviating: Deviating restarts Phase I_i, gaining at most M in one period but adding L more punishment periods. Compliance ends the punishment sooner and leads to w_i^i forever. For large δ, the cost of extending punishment exceeds any one-period gain.

Since all deviations are unprofitable for $\delta > \underline{\delta}$ (where $\underline{\delta}$ depends on the game parameters $M, v, w^i, \underline{v}_i^{\text{mix}}, L$), the constructed strategy profile is a subgame perfect equilibrium. □

Remark 4.1. The full dimensionality condition is needed only for $n \geq 3$ players: it allows constructing punishment payoffs w^i that reward the punishers while punishing the deviator. For two-player games, Fudenberg and Maskin (1986) showed the folk theorem holds without this condition, since minmaxing the opponent is the only thing the punisher can do.

Subgame Perfect Folk Theorem – Interpretation

For patient enough players (large δ), the set of SPE payoffs of the infinitely repeated game is exactly the set of feasible, strictly individually rational payoff vectors (under the full dimensionality condition). This is a much stronger result than the Nash folk theorem because SPE requires credible (self-enforcing) punishments.

Folk Theorems and Multi-Agent RL

The folk theorems have profound implications for MARL:

(i) **Multiplicity of equilibria:** In repeated general-sum games, there are typically *many* equilibria. This means that independent learners may converge to vastly different outcomes depending on initial conditions, learning rates, and exploration schedules.

(ii) **Emergent cooperation:** The folk theorems show that cooperation can be sustained by punishment threats, explaining why self-play agents sometimes learn cooperative strategies in iterated social dilemmas.

(iii) **Reward shaping:** The discount factor δ (analogous to γ in RL) determines the set of achievable equilibria. Curriculum designers can influence emergent behavior by adjusting γ.

5 Zero-Sum Games: Fictitious Play and No-Regret Learning

We return to two-player zero-sum games $G = (\{1,2\}, A_1, A_2, g)$ where $g_1(a) = -g_2(a) = g(a)$. Recall from Lesson 11 the minimax theorem: $\max_{\alpha_1} \min_{\alpha_2} \mathbb{E}[g(\alpha_1, \alpha_2)] = \min_{\alpha_2} \max_{\alpha_1} \mathbb{E}[g(\alpha_1, \alpha_2)] = v^*$ (the value of the game).

5.1 Fictitious Play

Definition 5.1 (Fictitious Play)**.** *Fictitious play* is a learning dynamic where each player best-responds to the empirical frequency of the opponent's past actions. Formally, let $a_j^1, a_j^2, \ldots$ be the actions played by player j. The empirical frequency after T rounds is

$$\hat{\alpha}_j^T(a_j) = \frac{1}{T} \sum_{t=1}^{T} \mathbf{1}\{a_j^t = a_j\} \qquad \text{for each } a_j \in A_j.$$

At round $T+1$, player i plays $a_i^{T+1} \in \mathrm{BR}_i(\hat{\alpha}_j^T)$, a best response to the empirical distribution of the opponent.

Theorem 5.1 (Convergence of Fictitious Play in 2×2 Zero-Sum Games)**.** *In any 2×2 zero-sum game, the empirical frequencies $(\hat{\alpha}_1^T, \hat{\alpha}_2^T)$ under fictitious play converge to a Nash equilibrium as $T \to \infty$.*

Proof. Let the payoff matrix be

$$G = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

with player 1 choosing rows and player 2 choosing columns, so $g_1 = G$, $g_2 = -G$.

Consider the function $\Phi^T = \max_{a_1} \sum_{t=1}^{T} g(a_1, a_2^t)$. This is player 1's "fictitious play payoff"—the best she could have done against the sequence of player 2's actions if constrained to a fixed action.

We analyze how Φ^T evolves. Note that $\Phi^T = \max\{R_1^T, R_2^T\}$ where $R_k^T = \sum_{t=1}^{T} g(k, a_2^t)$ is the cumulative payoff of row k.

In a 2×2 game, the best response of player 1 to $\hat{\alpha}_2^T$ is row 1 if $R_1^T > R_2^T$ and row 2 if $R_1^T < R_2^T$ (with either at ties). Define $D^T = R_1^T - R_2^T$.

Claim: $|D^T|$ is bounded. To see this, note that when $D^T > 0$, player 1 plays row 1, and player 2 best-responds to the empirical frequency $\hat{\alpha}_1^T$ which puts more weight on row 1. Player 2's best response tends to be the column that makes row 1 less attractive, which decreases D^{T+1}. Similarly when $D^T < 0$. The key observation in the 2×2 case is that D^T oscillates around 0 with bounded amplitude (bounded by $\max |g(a)|$), because each step changes D^T by at most $\max_{a_2} |g(1, a_2) - g(2, a_2)|$.

More precisely, define $V_1^T = \frac{1}{T}\Phi^T$ (player 1's "best fixed action" average payoff). We have

$$V_1^T = \frac{1}{T} \max_{a_1} \sum_{t=1}^{T} g(a_1, a_2^t) \geq \frac{1}{T} \sum_{t=1}^{T} g(a_1^t, a_2^t),$$

since the best fixed action does at least as well as the actual sequence.

Similarly, define $V_2^T = \frac{1}{T} \max_{a_2} \sum_{t=1}^{T} (-g(a_1^t, a_2))$ for player 2. By the minimax theorem for the stage game:

$$V_1^T \geq v^* \geq -V_2^T.$$

Furthermore, $V_1^T - v^* \to 0$ and $V_2^T + v^* \to 0$ because the "regret" $V_1^T - \frac{1}{T}\sum_t g(a_1^t, a_2^t)$ can be bounded by the bounded oscillation of $D^T/T \to 0$.

Since $|D^T|$ is bounded, $\frac{D^T}{T} \to 0$, which means $\frac{R_1^T}{T}$ and $\frac{R_2^T}{T}$ converge to the same limit. This implies that $\hat{\alpha}_2^T$ converges to making player 1 indifferent between rows, which is exactly the

NE condition in a 2×2 game with a completely mixed equilibrium. If the game has a pure equilibrium, fictitious play converges to it in finitely many steps.

Thus $(\hat{\alpha}_1^T, \hat{\alpha}_2^T) \to (\alpha_1^*, \alpha_2^*)$, a Nash equilibrium. □

Theorem 5.2 (Robinson's Theorem: Fictitious Play in General Zero-Sum Games)**.** *In any finite two-player zero-sum game, the empirical frequencies under fictitious play converge to the set of Nash equilibria.*

Proof sketch. The full proof (Robinson, 1951) uses a potential function argument. Define

$$\Phi_1^T = \max_{a_1 \in A_1} \sum_{t=1}^{T} g(a_1, a_2^t), \qquad \Phi_2^T = \max_{a_2 \in A_2} \sum_{t=1}^{T} (-g(a_1^t, a_2)).$$

Then $\frac{1}{T}\Phi_1^T \geq v^*$ and $\frac{1}{T}\Phi_2^T \geq -v^*$ (each player's best fixed action does at least as well as the game value against the empirical opponent distribution).

The key step is showing that $\frac{1}{T}(\Phi_1^T + \Phi_2^T) \to 0$. This follows because $\Phi_1^T + \Phi_2^T$ grows sublinearly: when player 1 plays a best response to $\hat{\alpha}_2^T$, the increase in $\Phi_1^{T+1} - \Phi_1^T$ is exactly $g(a_1^{T+1}, a_2^{T+1})$, while the increase $\Phi_2^{T+1} - \Phi_2^T \leq -g(a_1^{T+1}, a_2^{T+1}) + C$ for a constant C related to the number of actions. Carefully tracking these increments shows $\Phi_1^T + \Phi_2^T = O(\sqrt{T})$.

Since $\frac{1}{T}\Phi_1^T \to v^*$ and $\frac{1}{T}\Phi_2^T \to -v^*$, any limit point (α_1^*, α_2^*) of $(\hat{\alpha}_1^T, \hat{\alpha}_2^T)$ satisfies the NE conditions: α_1^* is a best response to α_2^* and vice versa. □

Fictitious Play and Policy-Space Response Oracles (PSRO)

Fictitious play is a precursor to the *PSRO* framework in modern MARL. In PSRO, each player maintains a population of policies (analogous to the empirical frequency in FP), and computes a best response to the current mixture—exactly as in fictitious play, but with neural-network policies instead of tabular strategies. The convergence guarantees of FP in zero-sum games provide theoretical backing for PSRO's effectiveness in competitive MARL.

5.2 No-Regret Learning

Definition 5.2 (External Regret)**.** Player i's *external regret* after T rounds is

$$R_i^T = \max_{a_i \in A_i} \sum_{t=1}^{T} g_i(a_i, a_{-i}^t) - \sum_{t=1}^{T} g_i(a_i^t, a_{-i}^t).$$

A learning algorithm has *no external regret* (or is *Hannan consistent*) if $R_i^T/T \to 0$ as $T \to \infty$.

Definition 5.3 (Multiplicative Weights Update (MWU))**.** Let $\eta > 0$ be a learning rate. Player i maintains weights $w_i^t(a_i)$ for each action $a_i \in A_i$:

$$w_i^1(a_i) = 1 \quad \text{for all } a_i,$$
$$w_i^{t+1}(a_i) = w_i^t(a_i) \cdot \exp\big(\eta \cdot g_i(a_i, a_{-i}^t)\big).$$

The mixed strategy at time t is $\alpha_i^t(a_i) = w_i^t(a_i) / \sum_{a_i'} w_i^t(a_i')$.

Theorem 5.3 (MWU Regret Bound)**.** *If $|g_i(a)| \leq 1$ for all a, then with $\eta = \sqrt{\frac{\ln |A_i|}{T}}$, the multiplicative weights update achieves*

$$R_i^T \leq 2\sqrt{T \ln |A_i|}.$$

In particular, $R_i^T/T \to 0$, so MWU is a no-regret algorithm.

Proof. Define the potential $W^t = \sum_{a_i \in A_i} w_i^t(a_i)$, so $W^1 = |A_i|$.

Upper bound on $\ln(W^{T+1}/W^1)$**:**

$$\begin{aligned}\frac{W^{t+1}}{W^t} &= \frac{\sum_{a_i} w_i^t(a_i)\exp(\eta \cdot g_i(a_i, a_{-i}^t))}{W^t} \\ &= \sum_{a_i} \alpha_i^t(a_i)\exp(\eta \cdot g_i(a_i, a_{-i}^t)) \\ &\le \sum_{a_i} \alpha_i^t(a_i)\big(1 + \eta g_i(a_i, a_{-i}^t) + \eta^2 g_i(a_i, a_{-i}^t)^2\big),\end{aligned}$$

using $e^x \le 1 + x + x^2$ for $|x| \le 1$ (valid when $\eta \le 1$ and $|g_i| \le 1$). Therefore:

$$\begin{aligned}\frac{W^{t+1}}{W^t} &\le 1 + \eta \sum_{a_i} \alpha_i^t(a_i) g_i(a_i, a_{-i}^t) + \eta^2 \\ &\le \exp\big(\eta \cdot g_i(\alpha_i^t, a_{-i}^t) + \eta^2\big),\end{aligned}$$

where $g_i(\alpha_i^t, a_{-i}^t) = \mathbb{E}_{a_i \sim \alpha_i^t}[g_i(a_i, a_{-i}^t)]$ and we used $1 + x \le e^x$. Telescoping:

$$\ln \frac{W^{T+1}}{W^1} \le \eta \sum_{t=1}^{T} g_i(\alpha_i^t, a_{-i}^t) + \eta^2 T.$$

Lower bound on $\ln(W^{T+1}/W^1)$**:** For any fixed action a_i^*:

$$W^{T+1} \ge w_i^{T+1}(a_i^*) = \exp\Big(\eta \sum_{t=1}^{T} g_i(a_i^*, a_{-i}^t)\Big).$$

Therefore:

$$\ln \frac{W^{T+1}}{W^1} \ge \eta \sum_{t=1}^{T} g_i(a_i^*, a_{-i}^t) - \ln |A_i|.$$

Combining:

$$\eta \sum_{t=1}^{T} g_i(a_i^*, a_{-i}^t) - \ln |A_i| \le \eta \sum_{t=1}^{T} g_i(\alpha_i^t, a_{-i}^t) + \eta^2 T.$$

Rearranging:

$$\sum_{t=1}^{T} g_i(a_i^*, a_{-i}^t) - \sum_{t=1}^{T} g_i(\alpha_i^t, a_{-i}^t) \le \frac{\ln |A_i|}{\eta} + \eta T.$$

Since this holds for all a_i^*, taking the max and substituting $\eta = \sqrt{\frac{\ln |A_i|}{T}}$:

$$R_i^T \le \sqrt{T \ln |A_i|} + \sqrt{T \ln |A_i|} = 2\sqrt{T \ln |A_i|}.$$ □

Theorem 5.4 (No-Regret Implies NE in Zero-Sum Games)**.** *In a two-player zero-sum game, if both players use no-regret learning algorithms, then the time-averaged strategy profile* $(\bar{\alpha}_1^T, \bar{\alpha}_2^T)$ *converges to the set of Nash equilibria as* $T \to \infty$*, where* $\bar{\alpha}_i^T = \frac{1}{T}\sum_{t=1}^{T} \alpha_i^t$*.*

Proof. Let $g = g_1 = -g_2$. The no-regret condition for player 1 gives:

$$\max_{a_1} \frac{1}{T} \sum_{t=1}^{T} g(a_1, \alpha_2^t) \le \frac{1}{T} \sum_{t=1}^{T} g(\alpha_1^t, \alpha_2^t) + \frac{R_1^T}{T}.$$

By linearity of g in the first argument and Jensen's inequality (or directly by linearity):

$$\max_{a_1} g(a_1, \bar{\alpha}_2^T) = \max_{a_1} \frac{1}{T} \sum_{t=1}^{T} g(a_1, \alpha_2^t) \leq \frac{1}{T} \sum_{t=1}^{T} g(\alpha_1^t, \alpha_2^t) + \varepsilon_1^T,$$

where $\varepsilon_1^T = R_1^T/T \to 0$.

Similarly, the no-regret condition for player 2 (who minimizes g) gives:

$$\min_{a_2} g(\bar{\alpha}_1^T, a_2) = \min_{a_2} \frac{1}{T} \sum_{t=1}^{T} g(\alpha_1^t, a_2) \geq \frac{1}{T} \sum_{t=1}^{T} g(\alpha_1^t, \alpha_2^t) - \varepsilon_2^T,$$

where $\varepsilon_2^T \to 0$.

Combining:

$$\min_{a_2} g(\bar{\alpha}_1^T, a_2) \geq \frac{1}{T} \sum_{t} g(\alpha_1^t, \alpha_2^t) - \varepsilon_2^T \geq \max_{a_1} g(a_1, \bar{\alpha}_2^T) - \varepsilon_1^T - \varepsilon_2^T.$$

But by the minimax inequality (which always holds):

$$\max_{a_1} g(a_1, \bar{\alpha}_2^T) \geq \min_{a_2} g(\bar{\alpha}_1^T, a_2).$$

Therefore:

$$0 \leq \max_{a_1} g(a_1, \bar{\alpha}_2^T) - \min_{a_2} g(\bar{\alpha}_1^T, a_2) \leq \varepsilon_1^T + \varepsilon_2^T \to 0.$$

This means the "duality gap" converges to zero. Since $\max_{a_1} g(a_1, \bar{\alpha}_2^T) \to v^*$ and $\min_{a_2} g(\bar{\alpha}_1^T, a_2) \to v^*$, any limit point (α_1^*, α_2^*) of $(\bar{\alpha}_1^T, \bar{\alpha}_2^T)$ satisfies $\max_{a_1} g(a_1, \alpha_2^*) = \min_{a_2} g(\alpha_1^*, a_2) = v^*$, which is the Nash equilibrium condition. □

No-Regret Learning in Multi-Agent RL

No-regret algorithms are fundamental to competitive MARL. The convergence guarantee of Theorem 5.4 underpins algorithms like:

- **Counterfactual Regret Minimization (CFR)** for extensive-form games (poker AI).
- **Online mirror descent** in policy space for continuous games.
- **V-learning** and other model-free algorithms that achieve sublinear regret against adversarial opponents.

The $O(\sqrt{T})$ regret rate of MWU translates to $O(1/\sqrt{T})$ convergence to NE in zero-sum games, which can be accelerated by optimistic variants.

6 Bayesian Games (Incomplete Information)

6.1 Motivation and Model

In many strategic settings, players are uncertain about others' payoff functions. A firm does not know a competitor's cost structure; an agent does not know an opponent's reward function. Harsanyi's (1967–68) framework models this via *types*.

Definition 6.1 (Bayesian Game)**.** A *Bayesian game* (or game of incomplete information) is a tuple $\Gamma = (N, (A_i)_{i \in N}, (\Theta_i)_{i \in N}, p, (u_i)_{i \in N})$ where:

(a) $N = [n]$ is the player set.

(b) A_i is the finite action set of player i.

(c) Θ_i is the finite *type space* of player i. A type $\theta_i \in \Theta_i$ encodes player i's private information.

(d) $p \in \Delta(\Theta)$ is the *common prior* distribution over the type profile $\theta = (\theta_1, \ldots, \theta_n) \in \Theta = \prod_i \Theta_i$.

(e) $u_i : A \times \Theta \to \mathbb{R}$ is player i's payoff function, depending on both the action profile and the type profile.

The timing is: Nature draws $\theta \sim p$; each player i observes only her own type θ_i; players simultaneously choose actions.

Definition 6.2 (Bayesian Nash Equilibrium)**.** A *(pure) strategy* in a Bayesian game is a function $s_i : \Theta_i \to A_i$ mapping types to actions. A strategy profile $s^* = (s_1^*, \ldots, s_n^*)$ is a *Bayesian Nash equilibrium* (BNE) if, for every player i and every type $\theta_i \in \Theta_i$:

$$s_i^*(\theta_i) \in \operatorname{argmax}_{a_i \in A_i} \mathbb{E}_{\theta_{-i} \sim p(\cdot|\theta_i)} \big[u_i\big(a_i, s_{-i}^*(\theta_{-i}), \theta_i, \theta_{-i}\big) \big],$$

where $p(\theta_{-i}|\theta_i) = p(\theta_i, \theta_{-i})/p(\theta_i)$ is the conditional distribution of others' types given θ_i.

Remark 6.1. The BNE condition says: each type of each player best-responds to the expected play of opponents, where the expectation is over opponents' types (which determine their actions through the strategy profile).

6.2 Existence of BNE

Theorem 6.1 (Existence of BNE)**.** *Every finite Bayesian game (finite player set, finite action sets, finite type spaces) has a Bayesian Nash equilibrium in mixed strategies.*

Proof. Transform the Bayesian game into the "agent normal form." For each player-type pair (i, θ_i), create an "agent" who chooses action $a_i \in A_i$. This yields a finite normal-form game with $\sum_i |\Theta_i|$ agents, each with action set A_i. The payoff to agent (i, θ_i) when the action profile is $(a_{j,\theta_j})_{j,\theta_j}$ is:

$$\sum_{\theta_{-i}} p(\theta_{-i}|\theta_i) \cdot u_i(a_{i,\theta_i}, (a_{j,\theta_j})_{j \neq i}, \theta_i, \theta_{-i}).$$

This is a finite normal-form game, so by Nash's theorem (from Lesson 11), it has a Nash equilibrium in mixed strategies. Any NE of the agent normal form corresponds to a BNE of the original Bayesian game: the mixed strategy of agent (i, θ_i) specifies $s_i(\theta_i) \in \Delta(A_i)$, and the NE condition for each agent is exactly the BNE condition for type θ_i of player i. □

6.3 Examples

Example 6.1 (Cournot Duopoly with Unknown Costs)**.** Two firms produce a homogeneous good with inverse demand $P(Q) = a - Q$ where $Q = q_1 + q_2$. Firm i's cost is $c_i q_i$. Suppose c_1 is known to both firms, but c_2 is private information of firm 2: $c_2 \in \{c_L, c_H\}$ with $\mathbb{P}(c_2 = c_L) = \pi$.

This is a Bayesian game with $\Theta_2 = \{c_L, c_H\}$, $\Theta_1 = \{c_1\}$ (trivial type space for firm 1). Actions are quantities $q_i \geq 0$.

Firm 2's profit when type is $\theta_2 = c$: $u_2 = (a - q_1 - q_2 - c)q_2$.

In a BNE, firm 2 uses strategy $s_2 : \{c_L, c_H\} \to \mathbb{R}_+$ and firm 1 chooses q_1. The first-order conditions give:

$$\begin{aligned} q_1 &= \frac{a - c_1 - \pi s_2(c_L) - (1-\pi)s_2(c_H)}{2}, \\ s_2(c_L) &= \frac{a - c_L - q_1}{2}, \\ s_2(c_H) &= \frac{a - c_H - q_1}{2}. \end{aligned}$$

Solving this system of three equations yields the unique BNE. The key insight is that firm 1's optimal quantity depends on the *expected* cost of firm 2, and firm 2's strategy conditions on its private type.

Example 6.2 (First-Price Sealed-Bid Auction)**.** Two bidders have private values $v_i \sim \text{Uniform}[0,1]$ independently. Each bidder submits a sealed bid b_i; the highest bidder wins and pays her bid. Payoff: $u_i = (v_i - b_i)\mathbf{1}\{b_i > b_j\}$.

This is a Bayesian game with $\Theta_i = [0,1]$ (continuous types). We seek a symmetric BNE where $b_i = \beta(v_i)$ for some increasing function β.

Given opponent's strategy β, bidder i with value v_i solves:

$$\max_b (v_i - b) \cdot \mathbb{P}(\beta(v_j) < b) = (v_i - b) \cdot \beta^{-1}(b).$$

The first-order condition (substituting $b = \beta(v_i)$) and the boundary condition $\beta(0) = 0$ yield $\beta(v_i) = v_i/2$: each bidder bids half her value.

Verification: If bidder j bids $v_j/2$, then bidder i's problem is $\max_b (v_i - b) \cdot 2b = \max_b 2bv_i - 2b^2$. The FOC gives $2v_i - 4b = 0$, so $b = v_i/2$, confirming the BNE.

Bayesian Games and Opponent Modeling in MARL

In MARL, agents typically do not know each other's reward functions—precisely the setting of Bayesian games. The Bayesian framework formalizes *opponent modeling*:

- The type θ_j can encode opponent j's policy parameters or reward weights.
- The prior $p(\theta_{-i})$ represents agent i's beliefs about opponents.
- Bayesian updating (observing opponents' actions and updating beliefs via Bayes' rule) is the principled approach to learning about opponents.

Algorithms like LOLA (Learning with Opponent-Learning Awareness) and Bayesian policy reuse implement variants of this Bayesian framework.

7 Perfect Bayesian Equilibrium

7.1 Motivation

In dynamic games of incomplete information, the BNE concept is too weak: it does not constrain behavior at off-equilibrium information sets. We need a refinement that combines sequential rationality with belief consistency.

Definition 7.1 (System of Beliefs)**.** In an extensive-form game with incomplete information, a *system of beliefs* μ assigns to each information set h a probability distribution $\mu(\cdot|h)$ over the nodes in that information set.

Definition 7.2 (Perfect Bayesian Equilibrium)**.** A *perfect Bayesian equilibrium* (PBE) is a pair (σ, μ) of a behavioral strategy profile σ and a belief system μ such that:

(a) **Sequential rationality:** At every information set h of every player i, the strategy σ_i is optimal given beliefs $\mu(\cdot|h)$ and the continuation strategies σ_{-i}.

(b) **Belief consistency:** At every information set h that is reached with positive probability under σ, the beliefs $\mu(\cdot|h)$ are derived from σ using Bayes' rule:

$$\mu(x|h) = \frac{\mathbb{P}(x \text{ is reached}|\sigma)}{\sum_{x' \in h} \mathbb{P}(x' \text{ is reached}|\sigma)} \quad \text{for all } x \in h.$$

(c) At information sets reached with zero probability (off the equilibrium path), beliefs are unrestricted (any distribution over nodes in the information set is allowed).

7.2 Signaling Games

A particularly important class of dynamic Bayesian games is *signaling games*.

Definition 7.3 (Signaling Game)**.** A *signaling game* has two players:

- **Sender (S):** Has private type $\theta \in \Theta$, drawn from prior $p(\theta)$. Observes θ, then sends a message $m \in M$.
- **Receiver (R):** Observes m (but not θ), then takes action $a \in A$.

Payoffs $u_S(\theta, m, a)$ and $u_R(\theta, m, a)$ depend on all three: type, message, and action.

Definition 7.4 (Equilibrium Types in Signaling Games)**.** (a) A PBE is *separating* if different types send different messages: $\sigma_S(\theta) \neq \sigma_S(\theta')$ for $\theta \neq \theta'$.

(b) A PBE is *pooling* if all types send the same message: $\sigma_S(\theta) = m^*$ for all θ.

(c) A PBE is *semi-separating* (or partially pooling) if some types pool and others separate.

Example 7.1 (Spence's Job Market Signaling)**.** A worker has ability $\theta \in \{L, H\}$ with $\mathbb{P}(\theta = H) = \pi$. The worker chooses education level $e \geq 0$ (the "signal"). An employer observes e and offers wage $w(e)$. Payoffs:

$$\begin{aligned} u_S(\theta, e, w) &= w - c(\theta, e), \quad \text{with } c(H, e) < c(L, e) \text{ for } e > 0, \\ u_R(\theta, e, w) &= \theta - w, \end{aligned}$$

where $c(\theta, e)$ is the cost of education (cheaper for high-ability workers: *single-crossing property*).

Separating equilibrium. In a separating PBE, $e_H \neq e_L$ and the receiver's beliefs at e_H are $\mu(H|e_H) = 1$, at e_L are $\mu(H|e_L) = 0$ (by Bayes' rule). The receiver offers $w(e_H) = H$ and $w(e_L) = L$.

For this to be an equilibrium, each type must prefer its own signal:

$$\begin{aligned} H - c(H, e_H) &\geq L - c(H, e_L) \quad (\text{type } H \text{ prefers } e_H), \\ L - c(L, e_L) &\geq H - c(L, e_H) \quad (\text{type } L \text{ prefers } e_L). \end{aligned}$$

The single-crossing property ensures that such e_H, e_L exist (typically $e_L = 0$ and e_H above a threshold).

Pooling equilibrium. Both types choose $e^* = 0$. The receiver's belief at e^* is the prior: $\mu(H|e^*) = \pi$, so $w(e^*) = \pi H + (1 - \pi)L$. Off-equilibrium beliefs at $e \neq e^*$ can be chosen to deter deviation, e.g., $\mu(H|e) = 0$ for all $e \neq e^*$, giving $w(e) = L$. Then no type wants to deviate if $\pi H + (1 - \pi)L - c(\theta, 0) \geq L - c(\theta, e)$ for all $e > 0$, which holds when the pooling wage is sufficiently attractive.

Proposition 7.1 (PBE Existence in Finite Signaling Games)**.** *Every finite signaling game (finite Θ, M, A) has a perfect Bayesian equilibrium.*

Proof. A finite signaling game is a finite extensive-form game. The sequential equilibrium existence theorem of Kreps and Wilson (1982) guarantees existence of a sequential equilibrium, which is a refinement of PBE. In particular, every sequential equilibrium is a PBE, so PBE exists.

Alternatively, one can construct a PBE directly: consider all possible sender strategies $\sigma_S : \Theta \to \Delta(M)$. For each such strategy, compute the receiver's beliefs via Bayes' rule at information sets reached with positive probability, and assign arbitrary beliefs at unreached information sets. The receiver's optimal response defines a best-response map. The sender's optimal message choice, given the receiver's response function, defines another best-response map. By Kakutani's fixed point theorem (or Nash's theorem applied to the agent normal form), a fixed point—hence a PBE—exists. □

8 Potential Games

8.1 Definitions

Potential games form a special class of games where all players' incentives can be captured by a single "potential function." This structure has powerful implications for existence of pure equilibria and convergence of learning dynamics.

Definition 8.1 (Exact Potential Game)**.** A finite game $G = (N, (A_i)_{i\in N}, (u_i)_{i\in N})$ is an *exact potential game* if there exists a function $\Phi : A \to \mathbb{R}$ (the *potential function*) such that for every player i, every $a_{-i} \in A_{-i}$, and every $a_i, a_i' \in A_i$:

$$u_i(a_i, a_{-i}) - u_i(a_i', a_{-i}) = \Phi(a_i, a_{-i}) - \Phi(a_i', a_{-i}).$$

Definition 8.2 (Ordinal Potential Game)**.** A finite game G is an *ordinal potential game* if there exists $\Phi : A \to \mathbb{R}$ such that for every player i, every a_{-i}, and every a_i, a_i':

$$u_i(a_i, a_{-i}) > u_i(a_i', a_{-i}) \iff \Phi(a_i, a_{-i}) > \Phi(a_i', a_{-i}).$$

Remark 8.1. Every exact potential game is an ordinal potential game. The converse is false: ordinal potentials only preserve the *sign* of payoff differences, not their magnitude.

8.2 Existence of Pure Nash Equilibria

Theorem 8.1 (Potential Games Have Pure NE)**.** *Every finite exact potential game (and every finite ordinal potential game) has at least one pure Nash equilibrium.*

Proof. Let $\Phi : A \to \mathbb{R}$ be the potential function. Since A is finite, Φ attains its maximum at some $a^* \in A$:

$$a^* \in \operatorname{argmax}_{a\in A} \Phi(a).$$

We claim a^* is a pure Nash equilibrium.

For any player i and any deviation $a_i' \in A_i$:

$$\Phi(a_i^*, a_{-i}^*) \geq \Phi(a_i', a_{-i}^*)$$

by the maximality of a^*. By the potential property:

$$u_i(a_i^*, a_{-i}^*) - u_i(a_i', a_{-i}^*) = \Phi(a_i^*, a_{-i}^*) - \Phi(a_i', a_{-i}^*) \geq 0.$$

Thus a_i^* is a best response to a_{-i}^* for every player i, so a^* is a NE.

For ordinal potential games, the same argument works: $\Phi(a_i^*, a_{-i}^*) \geq \Phi(a_i', a_{-i}^*)$ implies $u_i(a_i^*, a_{-i}^*) \geq u_i(a_i', a_{-i}^*)$ by the ordinal potential property (since if the strict inequality failed, i.e., $u_i(a_i', a_{-i}^*) > u_i(a_i^*, a_{-i}^*)$, then $\Phi(a_i', a_{-i}^*) > \Phi(a_i^*, a_{-i}^*)$, contradicting maximality). □

8.3 Convergence of Best-Response Dynamics

Definition 8.3 (Best-Response Dynamics)**.** In *(asynchronous) best-response dynamics*, players take turns updating their actions. At each step, one player i is selected (deterministically or randomly) and switches to a best response to the current actions of all other players:

$$a_i^{t+1} \in \operatorname{argmax}_{a_i \in A_i} u_i(a_i, a_{-i}^t), \qquad a_j^{t+1} = a_j^t \text{ for } j \neq i.$$

Theorem 8.2 (Finite Improvement Property)**.** *In every finite exact potential game, best-response dynamics converge to a pure Nash equilibrium in a finite number of steps.*

Proof. We show that Φ strictly increases at each step where a player's action changes, hence the dynamics must terminate (since A is finite and Φ cannot increase indefinitely).

Suppose at step t, player i switches from a_i^t to $a_i^{t+1} \neq a_i^t$ with $a_i^{t+1} \in \operatorname{argmax}_{a_i} u_i(a_i, a_{-i}^t)$. Since a_i^{t+1} is a strictly improving move (otherwise the player would not switch), we have:

$$u_i(a_i^{t+1}, a_{-i}^t) > u_i(a_i^t, a_{-i}^t).$$

By the exact potential property:

$$\Phi(a_i^{t+1}, a_{-i}^t) - \Phi(a_i^t, a_{-i}^t) = u_i(a_i^{t+1}, a_{-i}^t) - u_i(a_i^t, a_{-i}^t) > 0.$$

Thus $\Phi(a^{t+1}) > \Phi(a^t)$ whenever an action changes. Since A is finite, the sequence $\Phi(a^1), \Phi(a^2), \ldots$ is strictly increasing and takes values in the finite set $\{\Phi(a) : a \in A\}$, so it must terminate. Termination occurs when no player wants to change, i.e., at a pure Nash equilibrium.

Remark on step count: The number of steps is bounded by $|A| = \prod_i |A_i|$ (since each action profile can appear at most once in the improvement path), but this worst-case bound is typically very loose. □

Corollary 8.3. *The finite improvement property also holds for* better-response dynamics *(where player i switches to any action that improves her payoff, not necessarily the best response), since Φ still strictly increases.*

Proof. If $u_i(a_i', a_{-i}^t) > u_i(a_i^t, a_{-i}^t)$, then $\Phi(a_i', a_{-i}^t) > \Phi(a_i^t, a_{-i}^t)$ by the potential property. The same termination argument applies. □

8.4 Congestion Games

The most important class of potential games in applications is congestion games.

Definition 8.4 (Congestion Game)**.** A *congestion game* is defined by $(N, E, (A_i)_{i \in N}, (d_e)_{e \in E})$ where:

- $N = [n]$ is the player set.
- E is a finite set of *resources* (or *facilities*).
- $A_i \subseteq 2^E$ is a collection of subsets of resources available to player i. Each action $a_i \in A_i$ is a set of resources that player i uses.
- $d_e : \{1, \ldots, n\} \to \mathbb{R}$ is the *delay function* (or cost function) for resource e, depending on the number of users.

The payoff (negative cost) to player i under action profile $a = (a_1, \ldots, a_n)$ is:

$$u_i(a) = -\sum_{e \in a_i} d_e\big(n_e(a)\big),$$

where $n_e(a) = |\{j \in N : e \in a_j\}|$ is the congestion (number of users) of resource e.

Theorem 8.4 (Congestion Games are Exact Potential Games (Rosenthal))**.** *Every congestion game is an exact potential game with potential function*

$$\Phi(a) = -\sum_{e \in E} \sum_{k=1}^{n_e(a)} d_e(k).$$

Proof. We must verify that for any player i, any a_{-i}, and any $a_i, a_i' \in A_i$:

$$u_i(a_i, a_{-i}) - u_i(a_i', a_{-i}) = \Phi(a_i, a_{-i}) - \Phi(a_i', a_{-i}).$$

Consider the left side. Let $n_e = n_e(a_i, a_{-i})$ denote the congestion when player i uses a_i, and $n_e' = n_e(a_i', a_{-i})$ when player i uses a_i'. We partition the resources into three groups:

(i) $e \in a_i \cap a_i'$: used by i under both actions. Then $n_e = n_e'$, so $d_e(n_e)$ cancels in both u_i and Φ differences.

(ii) $e \in a_i \setminus a_i'$: used by i under a_i but not a_i'. Then $n_e = n_e' + 1$ (one more user under a_i). The contribution to $u_i(a_i, a_{-i}) - u_i(a_i', a_{-i})$ is $-d_e(n_e) = -d_e(n_e' + 1)$. The contribution to $\Phi(a_i, a_{-i}) - \Phi(a_i', a_{-i})$ is $-\sum_{k=1}^{n_e'+1} d_e(k) + \sum_{k=1}^{n_e'} d_e(k) = -d_e(n_e' + 1)$. These are equal.

(iii) $e \in a_i' \setminus a_i$: used by i under a_i' but not a_i. Then $n_e' = n_e + 1$. The contribution to $u_i(a_i, a_{-i}) - u_i(a_i', a_{-i})$ is $0 - (-d_e(n_e')) = d_e(n_e + 1)$. The contribution to $\Phi(a_i, a_{-i}) - \Phi(a_i', a_{-i})$ is $-\sum_{k=1}^{n_e} d_e(k) + \sum_{k=1}^{n_e+1} d_e(k) = d_e(n_e + 1)$. These are equal.

Summing over all resources, $u_i(a_i, a_{-i}) - u_i(a_i', a_{-i}) = \Phi(a_i, a_{-i}) - \Phi(a_i', a_{-i})$. □

Corollary 8.5. *Every finite congestion game has a pure Nash equilibrium, and best-response dynamics converge to one in finitely many steps.*

Proof. Immediate from Theorems 8.4, 8.1, and 8.2. □

Example 8.1 (Network Routing Game)**.** Consider a network with source s and destination d, and n users each choosing a path from s to d. Each edge e has delay $d_e(k)$ depending on congestion k. This is a congestion game: resources are edges, each player's action set is the set of s-d paths, and the total delay on a path is $\sum_{e \in \text{path}} d_e(n_e)$. By Corollary 8.5, a pure NE exists and can be found by best-response dynamics.

Example 8.2 (Resource Allocation)**.** Consider n agents each choosing one of m machines to process a job. Machine e has processing time $d_e(k) = k \cdot \ell_e$ (linear congestion) where ℓ_e is the base load. Each agent's cost is the processing time on her chosen machine. This is a congestion game with $A_i = E = \{1, \ldots, m\}$ (each action uses a single resource). The potential function is $\Phi(a) = -\sum_{e=1}^{m} \ell_e \cdot \frac{n_e(n_e+1)}{2}$, and pure NE exist with convergent best-response dynamics.

Potential Games and Independent Learning in MARL

Potential games are among the few game classes where independent learners (agents running separate RL algorithms without coordination) are guaranteed to converge:

- **Best-response dynamics** correspond to each agent independently computing optimal policies given fixed opponent policies. The finite improvement property ensures convergence.
- **Log-linear learning:** If each agent plays action a_i with probability proportional to $\exp(\beta \cdot u_i(a_i, a_{-i}))$ (Boltzmann exploration), the joint process is a Markov chain that converges to the potential maximizer as $\beta \to \infty$.

- **Multi-agent reward shaping:** If a team reward function Φ can serve as a potential for individual reward functions u_i, then shaping individual rewards via Φ makes the game a potential game, guaranteeing convergence.

This is the theoretical basis for *difference rewards* and *Wonderful Life Utility* in cooperative MARL.

9 Connections to Multi-Agent RL

We consolidate the connections between the game-theoretic results of this lesson and modern multi-agent reinforcement learning.

Independent Learners in Repeated Games

When n agents independently run Q-learning or policy gradient in a repeated game, each agent treats the others as part of the environment. The folk theorems (Theorems 4.1 and 4.2) imply that the set of stable outcomes is vast—any feasible, individually rational payoff can be a fixed point. This explains the well-known instability and sensitivity to hyperparameters of independent learners in general-sum games. Convergence guarantees for independent Q-learning exist only in restrictive settings: two-player zero-sum games (minimax-Q), potential games, and certain classes of team games.

Self-Play and Fictitious Play

Self-play—where an agent trains against copies of itself—is a form of fictitious play. In zero-sum games, Robinson's theorem (Theorem 5.2) guarantees that the empirical strategy profile converges to NE. Modern algorithms extend this:

- **Neural Fictitious Self-Play (NFSP):** Uses neural networks to approximate best responses and average strategies, converging to approximate NE in large zero-sum games.
- **PSRO (Policy-Space Response Oracles):** Maintains a population of policies and computes a best response to the meta-mixture, directly implementing fictitious play in policy space.
- **Double Oracle:** Iteratively adds best responses to a restricted game, converging to the NE of the full game.

The regret bound of MWU (Theorem 5.3) provides the convergence rate: $O(1/\sqrt{T})$ for last-iterate convergence in zero-sum games with optimistic updates.

Opponent Modeling as Bayesian Inference

The Bayesian game framework (Section 6) provides the foundation for opponent modeling in MARL:

(i) **Type inference:** An agent maintains a belief $p(\theta_{-i}|h^t)$ over opponent types, updated via Bayes' rule as actions are observed.

(ii) **Bayesian policy optimization:** The agent's policy maximizes expected return under the current belief: $\max_{\pi_i} \mathbb{E}_{\theta_{-i}\sim p(\cdot|h^t)}[R_i(\pi_i, \pi_{-i}(\theta_{-i}))]$.

(iii) **Exploration-exploitation:** The agent may take actions to learn about opponents' types (information-gathering exploration), connecting to the theory of Bayesian exploration in MDPs.

Potential-Based Reward Shaping in Multi-Agent Settings

Ng, Harada, and Russell's (1999) potential-based reward shaping theorem states that adding a shaping reward $F(s, s') = \gamma\phi(s') - \phi(s)$ preserves the optimal policy in single-agent MDPs. In multi-agent settings, this connects to potential games:

- If the joint reward function serves as a potential for individual rewards, the multi-agent system is a potential game and independent learning converges.
- The *difference reward* $D_i(a) = R(a) - R(a_{-i}, c_i)$ (where c_i is a default action for agent i) is designed so that $D_i(a_i, a_{-i}) - D_i(a'_i, a_{-i}) = R(a_i, a_{-i}) - R(a'_i, a_{-i})$, making R an exact potential for the game with payoffs D_i.
- This theoretical property ensures that maximizing individual difference rewards aligns with maximizing the global objective, providing a principled approach to cooperative MARL.

10 Exercises

Exercise 10.1 (Finitely Repeated Prisoner's Dilemma)**.** Consider the Prisoner's Dilemma from Example 3.1, repeated $T = 100$ times. Show that the unique SPE is mutual defection in every period, regardless of T. Explain why this result is "paradoxical" and how it motivates infinitely repeated games.

Exercise 10.2 (Grim Trigger in a Different Game)**.** Consider the stage game:

	L	C	R
T	$6,6$	$0,8$	$0,0$
M	$8,0$	$4,4$	$0,0$
B	$0,0$	$0,0$	$2,2$

(a) Find all pure Nash equilibria of the stage game.

(b) Compute the minimax values $\underline{v}_1$ and $\underline{v}_2$.

(c) Show that the payoff $(6, 6)$ (from (T, L)) can be sustained as an SPE of the infinitely repeated game using grim trigger with punishment (B, R). Find the minimum discount factor.

(d) Can the payoff $(7, 7)$ be sustained? If so, construct a strategy profile.

Exercise 10.3 (Tit-for-Tat Is Not SPE)**.** Show that in the infinitely repeated Prisoner's Dilemma (Example 3.1), the tit-for-tat strategy profile is a Nash equilibrium for $\delta \geq 1/3$ but is *not* a subgame perfect equilibrium. (Hint: Consider the subgame after (D, C).)

Exercise 10.4 (Computing BNE)**.** Consider a two-player Bayesian game where player 1 has type $\theta_1 \in \{H, L\}$ with $\mathbb{P}(\theta_1 = H) = 1/2$, and player 2 has no private information. Actions: $A_1 = A_2 = \{U, D\}$. Payoffs (for each type of player 1):

Type H :

	L	R
U	$2,1$	$0,0$
D	$0,0$	$1,2$

Type L :

	L	R
U	$0,0$	$1,2$
D	$2,1$	$0,0$

Find all Bayesian Nash equilibria (pure and mixed).

Exercise 10.5 (Second-Price Auction as Bayesian Game)**.** In a second-price sealed-bid auction with n bidders having independent private values $v_i \sim F$ (where F is a continuous CDF on $[0, \bar{v}]$), show that bidding $b_i = v_i$ (truthful bidding) is a dominant strategy, hence a BNE. Compare with the first-price auction BNE from Example 6.2.

Exercise 10.6 (Verifying a Potential Function)**.** Consider the two-player game with $A_1 = A_2 = \{X, Y\}$ and payoffs:

	X	Y
X	$3, 2$	$1, 1$
Y	$2, 1$	$4, 3$

(a) Show this is an exact potential game by constructing the potential function Φ.

(b) Find all pure Nash equilibria using Φ.

(c) Verify that best-response dynamics converge from any initial profile.

Exercise 10.7 (MWU Convergence Rate)**.** Consider the zero-sum game Rock-Paper-Scissors:

	R	P	S
R	0	-1	1
P	1	0	-1
S	-1	1	0

(a) What is the unique NE and the value of the game?

(b) If both players use MWU with $\eta = \sqrt{\ln 3/T}$, bound the duality gap $\max_{a_1} g(a_1, \bar{\alpha}_2^T) - \min_{a_2} g(\bar{\alpha}_1^T, a_2)$ after $T = 1000$ rounds.

(c) How many rounds are needed for the duality gap to be below 0.01?

Exercise 10.8 (Folk Theorem Application)**.** Consider the two-player stage game:

	C	D
C	$5, 5$	$0, 7$
D	$7, 0$	$2, 2$

(a) Compute the feasible payoff set V and the minimax values.

(b) Apply the Nash folk theorem to determine which payoff vectors are achievable as NE payoffs for large δ.

(c) For the payoff $(5, 5)$, find the minimum discount factor under grim trigger.

(d) For the payoff $(4, 4)$, describe a strategy profile achieving it as an SPE and find the minimum δ.

Exercise 10.9 (Congestion Game with Three Players)**.** Three players share two resources $\{A, B\}$. Each player chooses exactly one resource. Delay functions: $d_A(k) = k$ and $d_B(k) = 2k - 1$.

(a) Write the Rosenthal potential function Φ.

(b) Enumerate all action profiles and compute Φ for each.

(c) Find all pure NE by maximizing Φ (equivalently, minimizing $-\Phi$).

(d) Verify that your NE are indeed equilibria by checking no player wants to deviate.

Exercise 10.10 (Separating vs. Pooling Equilibria)**.** Consider a signaling game with two sender types $\theta \in \{G, B\}$ (good, bad), $\mathbb{P}(\theta = G) = 1/3$, two messages $m \in \{H, L\}$ (high, low effort), and two receiver actions $a \in \{A, R\}$ (accept, reject). Payoffs:

$$u_S(G, H, A) = 3, \quad u_S(G, H, R) = -1, \quad u_S(G, L, A) = 4, \quad u_S(G, L, R) = 0,$$
$$u_S(B, H, A) = 3, \quad u_S(B, H, R) = -2, \quad u_S(B, L, A) = 4, \quad u_S(B, L, R) = 1,$$
$$u_R(\theta, m, A) = \begin{cases} 1 & \text{if } \theta = G, \\ -1 & \text{if } \theta = B, \end{cases} \quad u_R(\theta, m, R) = 0.$$

(a) Does a separating PBE exist where type G sends H and type B sends L?

(b) Does a pooling PBE exist where both types send L?

(c) Does a pooling PBE exist where both types send H?

11 Summary of Key Results

Result	**Statement**	**Ref.**
Unraveling Theorem	Unique stage-game NE $\Rightarrow$ unique SPE of G^T: play NE every period	Thm 2.1
Nash Folk Theorem	Any feasible, strictly individually rational payoff is a NE payoff for δ near 1	Thm 4.1
SPE Folk Theorem	Same as Nash folk theorem but for SPE (under full dimensionality condition)	Thm 4.2
Fictitious Play (2×2)	Empirical frequencies converge to NE in 2×2 zero-sum games	Thm 5.1
Robinson's Theorem	Fictitious play converges to NE in all finite two-player zero-sum games	Thm 5.2
MWU Regret Bound	External regret $\leq 2\sqrt{T \ln \lvert A_i \rvert}$; hence MWU is no-regret	Thm 5.3
No-Regret $\Rightarrow$ NE	If both players use no-regret, time-avg. play converges to NE in zero-sum	Thm 5.4
BNE Existence	Every finite Bayesian game has a BNE in mixed strategies	Thm 6.1
PBE in Signaling Games	Every finite signaling game has a PBE	Prop 7.1
Potential $\Rightarrow$ Pure NE	Every finite potential game has a pure Nash equilibrium	Thm 8.1
Finite Improvement	Best-response dynamics converge in finite steps in potential games	Thm 8.2
Rosenthal's Theorem	Every congestion game is an exact potential game	Thm 8.4

Game Class	**MARL Implication**
Repeated games (general-sum)	Folk theorems $\Rightarrow$ many equilibria; independent learners may converge to any of them; sensitive to initialization and hyperparameters
Zero-sum games	Minimax + no-regret $\Rightarrow$ convergent self-play (NFSP, PSRO, CFR)
Bayesian games	Opponent modeling as Bayesian inference over types; principled exploration
Potential games	Independent best-response / Q-learning converges; difference rewards make cooperative MARL a potential game
Congestion games	Routing, resource allocation: pure NE exist, local search works